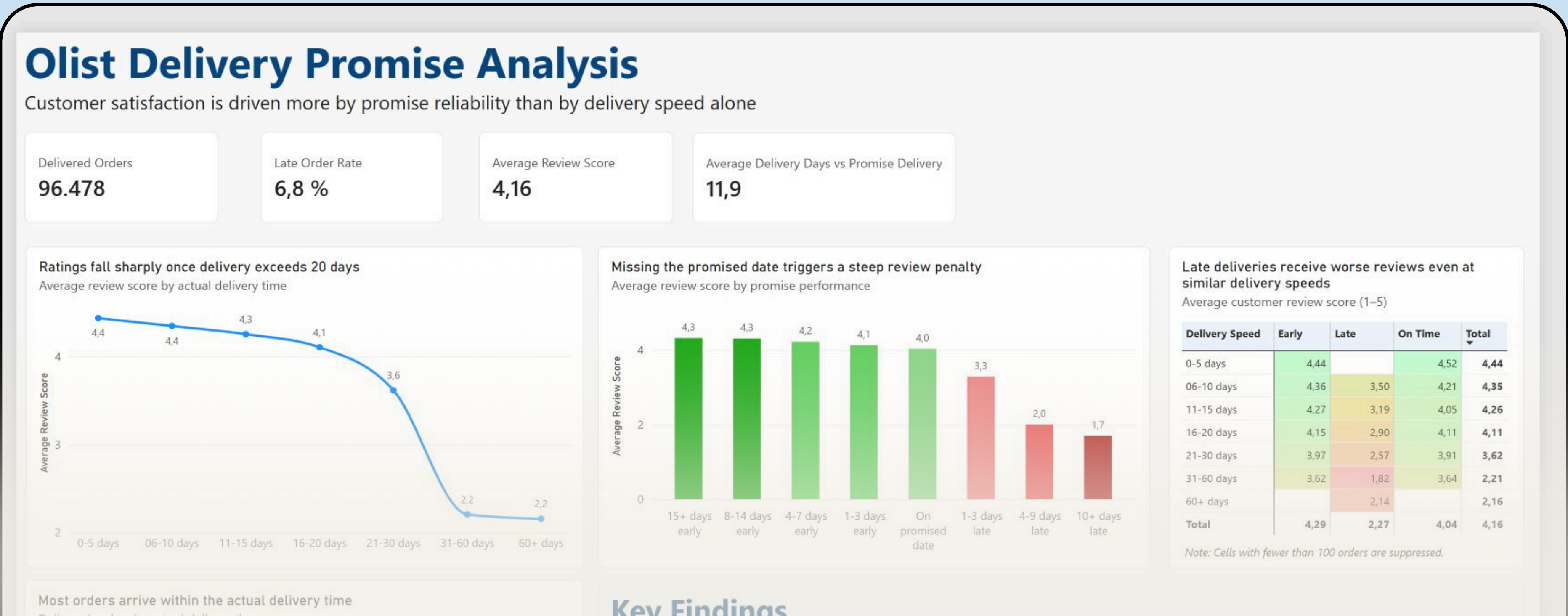


Olist delivery promise analysis

A Power BI analysis of delivery promise reliability and calibration across Brazilian states, using 96k order data.

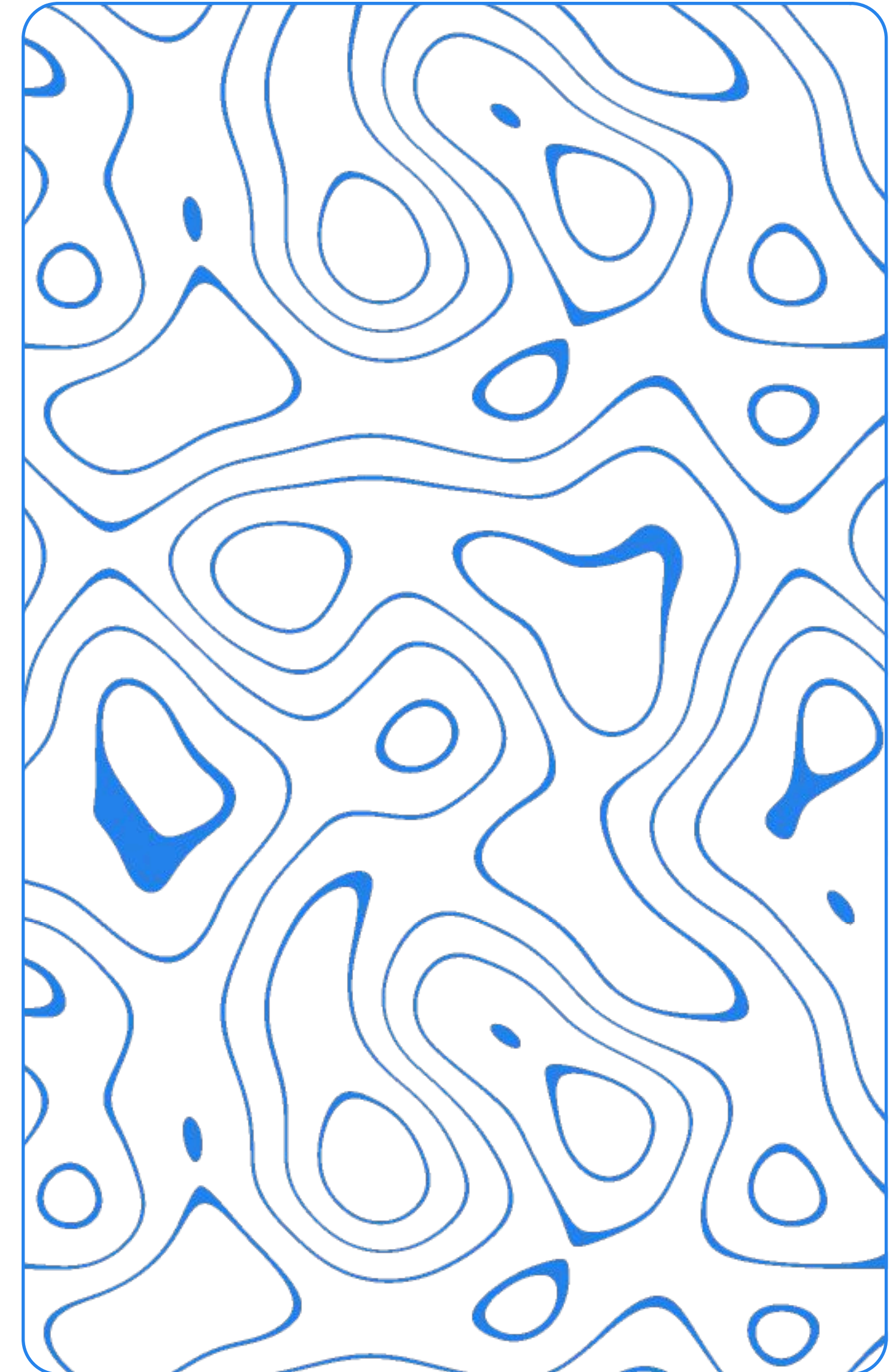


About the Project

Olist is a Brazilian e-commerce marketplace connecting thousands of sellers to customers across the country.

Since Olist operates in a large country with many different states and locations, delivery promises here matter a great amount for customer satisfaction.

Whether delivery promises actually reflect these regional differences is not obvious from the data. How promises are set is an operational decision that affects both customer satisfaction and the credibility of the marketplace.



The Questions

The analysis explores two central questions.

1. Is customer satisfaction driven by how fast an order arrives, or by whether it arrives when promised?
2. Across all states, does the delivery promise window scale with delivery variability, or only with delivery speed?

The two questions are connected since the first one establishes what is actually responsible for customer satisfaction. The second asks whether Olist's delivery promises are set in a way that accounts for it.

192,000

56%

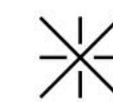
240

160

80

0

Feb



583.7

↓12.4%

Missing the promise costs more than being slow.

Olist Delivery Promise Analysis

Customer satisfaction is driven more by promise reliability than by delivery speed alone

Delivered Orders

96.478

Late Order Rate

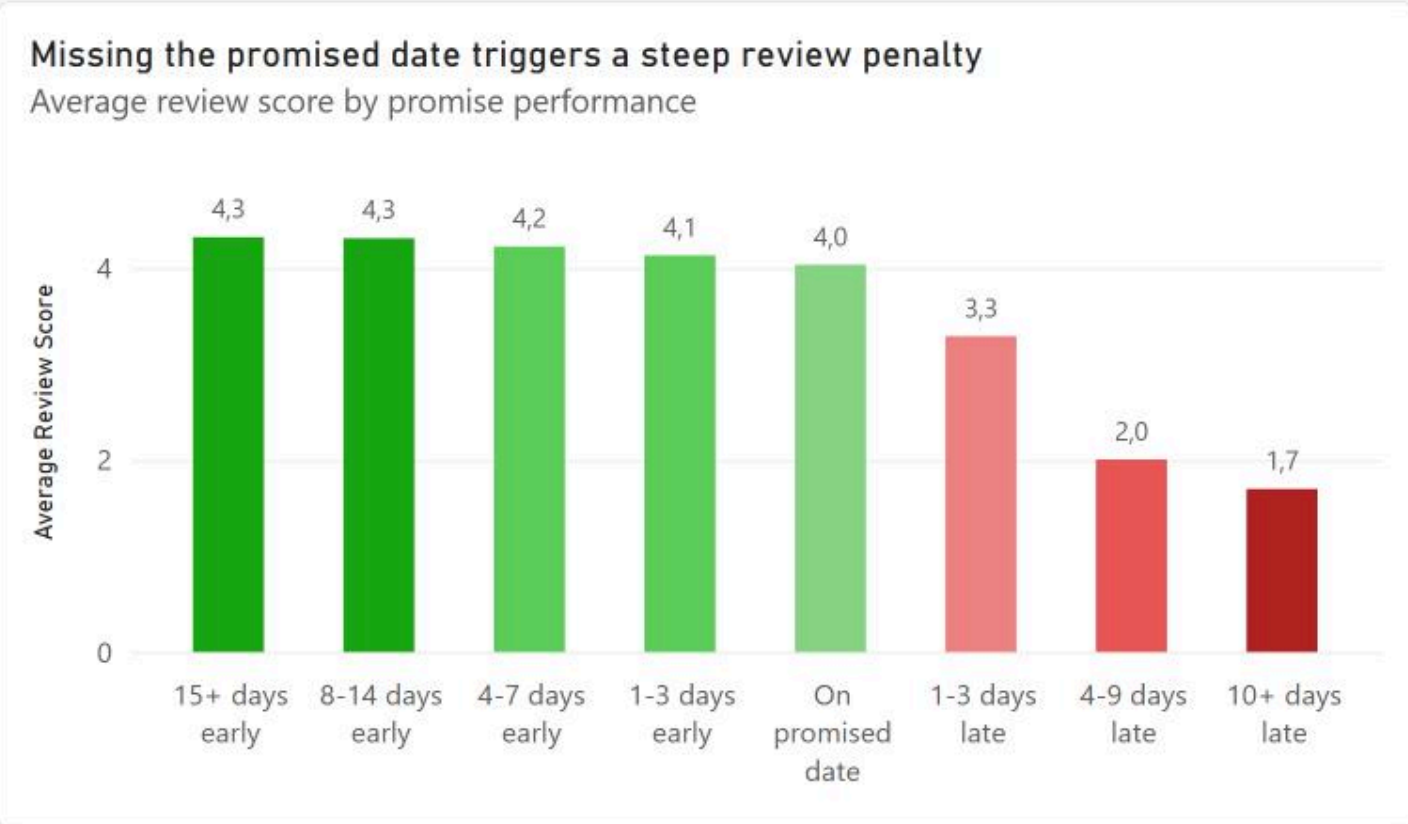
6,8 %

Average Review Score

4,16

Average Delivery Days vs Promise Delivery

11,9

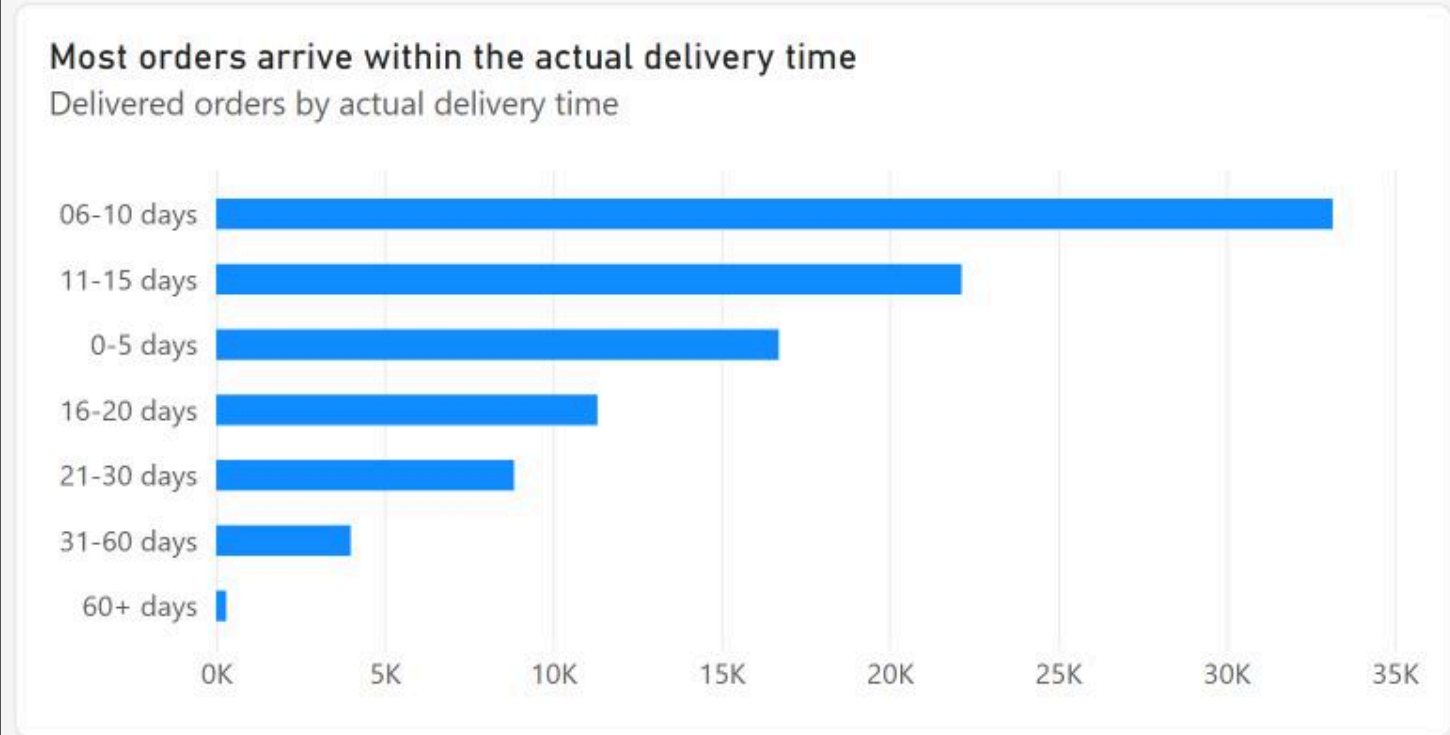


Late deliveries receive worse reviews even at similar delivery speeds

Average customer review score (1–5)

Delivery Speed	Early	Late	On Time	Total
0-5 days	4,44		4,52	4,44
06-10 days	4,36	3,50	4,21	4,35
11-15 days	4,27	3,19	4,05	4,26
16-20 days	4,15	2,90	4,11	4,11
21-30 days	3,97	2,57	3,91	3,62
31-60 days	3,62	1,82	3,64	2,21
60+ days		2,14		2,16
Total	4,29	2,27	4,04	4,16

Note: Cells with fewer than 100 orders are suppressed.



Key Findings

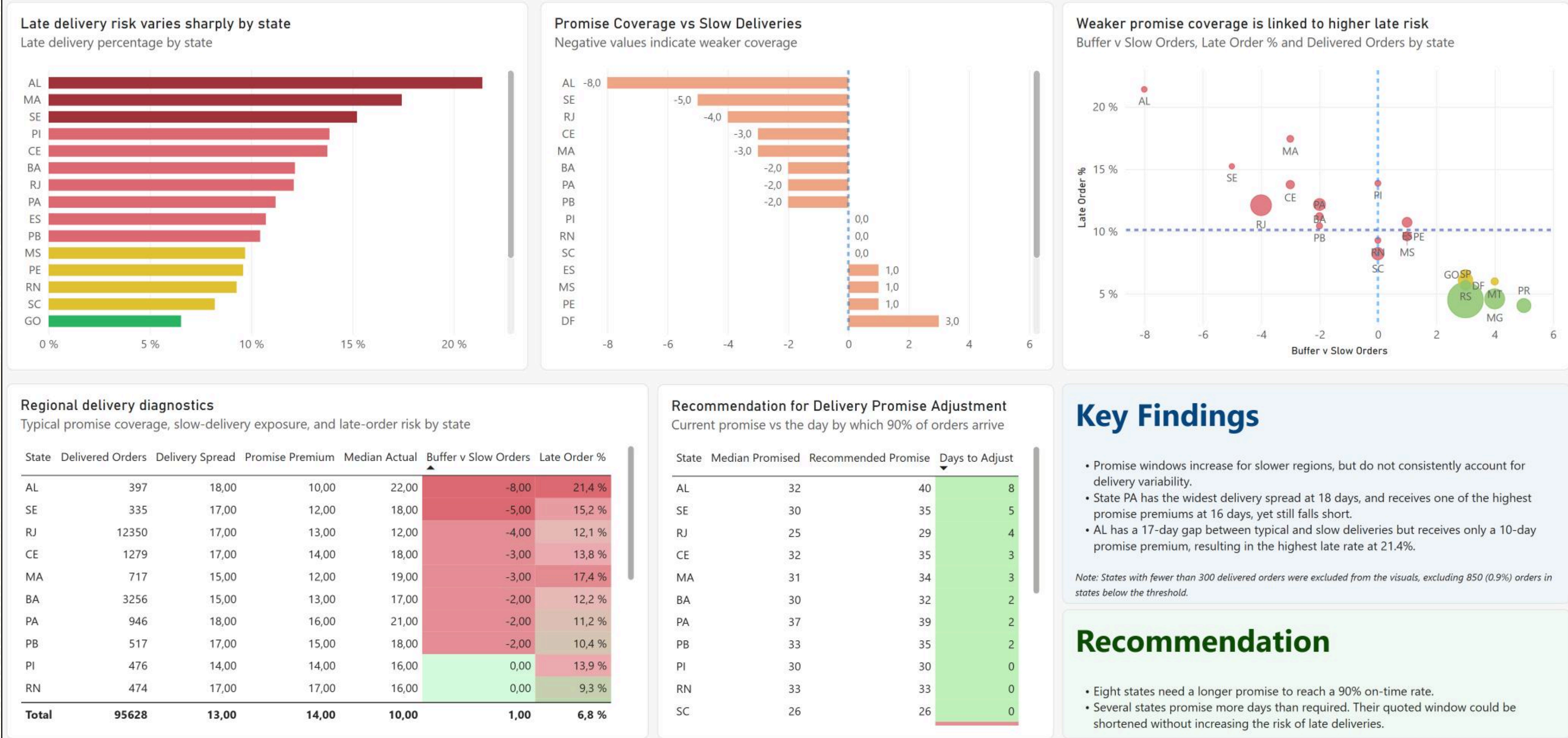
- Customers tolerate slower delivery reasonably well when the promised date is met.
- Satisfaction drops sharply when delivery promise is missed. This trend is seen even among orders with similar actual delivery times.
- The penalty for missing the promise grows with delivery time. Example: 0,86 points on a 6-10 day delivery, and 1,80 points on a 31-60 day delivery.
- Therefore, expectation management appears to be more important for customer satisfaction than actual delivery time.

Note: 8 delivered orders have no review and are excluded from review-score visuals.

Promises scale with speed, not with variability.

Regional Delivery Performance

Delivery promise windows track how slow a region is, but not how unpredictable it is.



Recommendation

Calibrate the promise to each state's slow orders

Late deliveries receive worse reviews even at similar delivery speeds

Average customer review score (1–5)

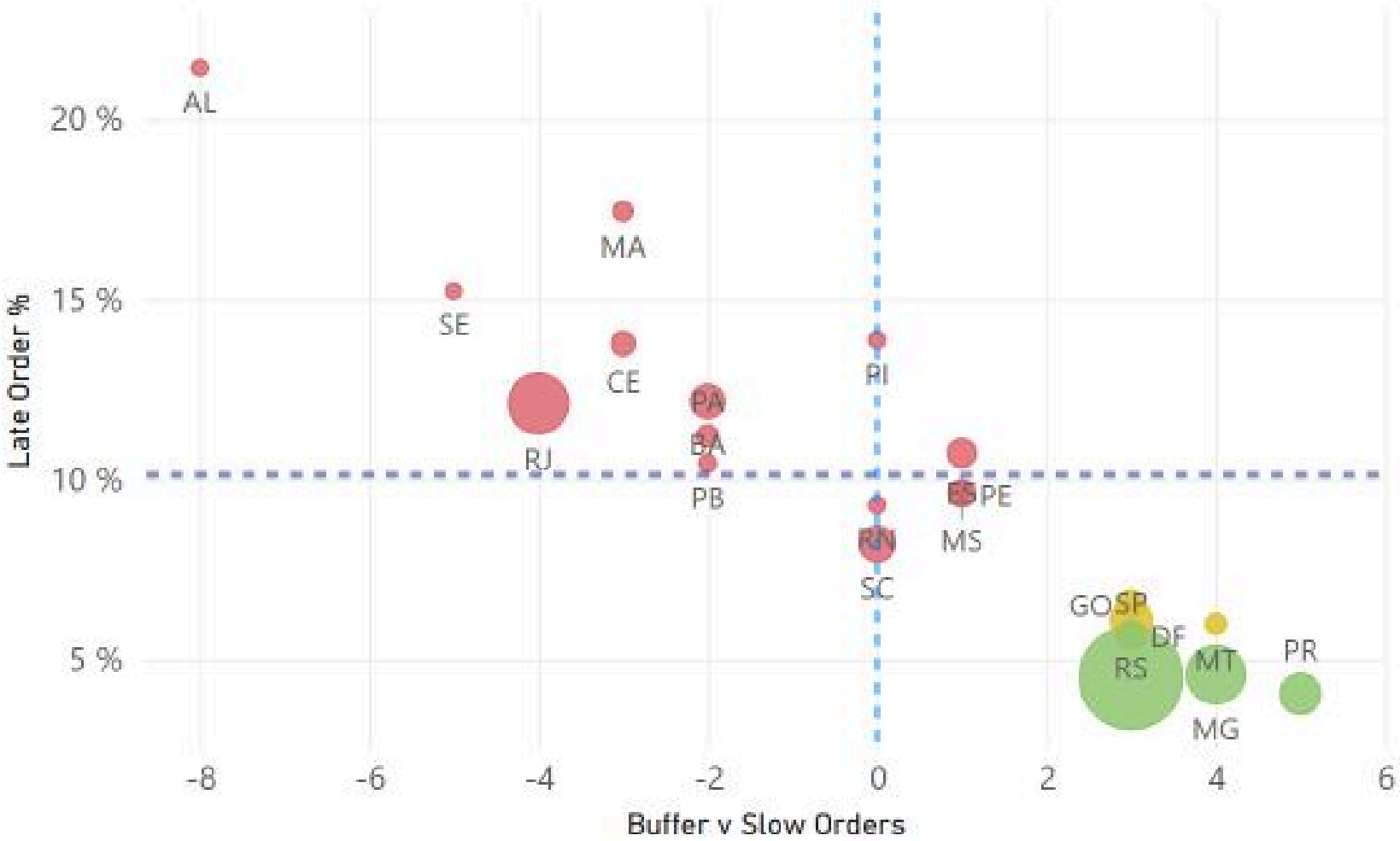
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Note: Cells with fewer than 100 orders are suppressed.

Among orders delivered in 6-10 days, late orders score 0.86 points lower than early ones. At 31-60 days, the gap widens to 1.80.

Weaker promise coverage is linked to higher late risk

Buffer v Slow Orders, Late Order % and Delivered Orders by state



States left of the centre line promise less time than a slow delivery requires, and they cluster above the 10% late-rate line.

Recommendation for Delivery Promise Adjustment

Current promise vs the day by which 90% of orders arrive

State	Median Promised	Recommended Promise	Days to Adjust
AL	32	40	8
SE	30	35	5
RJ	25	29	4
CE	32	35	3
MA	31	34	3
BA	30	32	2
PA	37	39	2
PB	33	35	2
PI	30	30	0
RN	33	33	0
SC	26	26	0
ES	26	25	-1
MS	26	25	-1
PE	32	31	-1
DF	25	22	-3
GO	28	25	-3
RS	29	26	-3
SP	19	16	-3
MG	24	20	-4
MT	32	28	-4
PR	25	20	-5

Eight states need a longer promise to reach a 90% on-time rate. Others promise more than required and could shorten their window.

The Problem

Delivery promise reliability drives satisfaction, and the promises are miscalibrated.

Recommendation

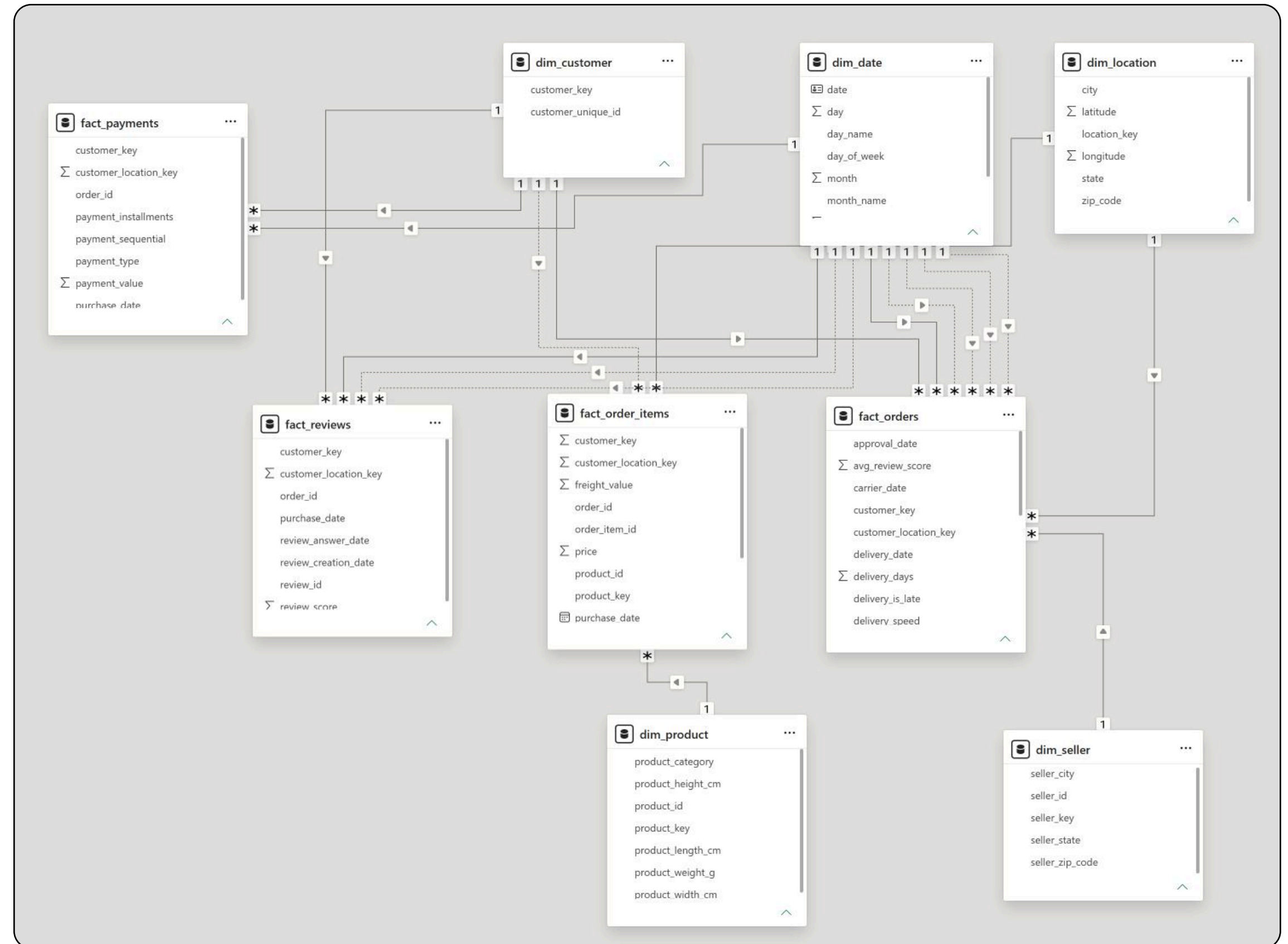
Set each state's delivery promise to the day by which 90% of its orders arrive, rather than adding a similar premium to slower regions. Eight states need a longer promise window, and ten could shorten theirs without increasing the risk of late deliveries.

The Data Model

Before



After



Nine source tables were flattened into a star schema with five dimensions and four fact tables.

- Flattened nine source tables into a star schema with five dimensions and four fact tables.
- Built dim_customer on customer_unique_id rather than customer_id, as customer_id is generated per order and does not represent a customer.
- Collapsed the geolocation table to one row per zip code. In a few cases a zip code appeared with multiple state values. They were mapped to the city and state pairing with the highest number of source records. This ensured that one zip code relates to the same city and state.
- Added a date dimension with role-playing relationships for purchase, approval, delivery, and estimated delivery dates.
- Replaced natural keys with surrogate keys across facts and dimensions.

The Data Model

After

Grain

Table	One row per
fact_orders	order
fact_order_items	item line within an order
fact_payments	payment sequential line within an order
fact_reviews	review-to-order assignment
dim_customer	customer
dim_seller	seller
dim_product	product
dim_location	zip code
dim_date	calendar date

Grain Notes

- fact_orders is an accumulating snapshot: one row per order, with date columns tracking purchase, approval, delivery, and the estimated delivery date.
- fact_reviews contains 99,224 rows against 98,673 distinct orders and 98,410 review IDs. Some orders carry multiple reviews, and some review records are attached to more than one order. Where an order had multiple reviews, the scores were averaged before being merged into fact_orders as avg_review_score.

Key Measures

Data

Search

_measure

Average Delivery Days vs Promise Delivery

Average Review Score

Average Review Score (Min 100 Orders)

Buffer v Slow Orders

Days to Adjust

Delivered Orders

Delivery Spread

Late Order %

Late Orders

Median Actual

Median Promised

Orders

Promise Premium

Recommended Promise

Average Review Score (Min 100 Orders)

The average review of orders for an order count of ≥ 100 , calculated from the measure "Average Review Score"

DAX

Average Review Score (Min 100 Orders) = IF([Orders] $\geq$ 100, [Average Review Score], BLANK())

Promise Premium

The median number of days Olist used as a buffer against the median of actual delivery days.

DAX

Promise Premium = [Median Promised] - [Median Actual]

Recommended Promise

The number of days it would require to successfully deliver 90% of a state's order, rounded to the next whole number.

DAX

Recommended Promise = CEILING(PERCENTILE.INC(fact_orders[delivery_days], 0.90),1)

Days to Adjust

The number of days needed to increase/decrease in order to match the recommended promise.

DAX

Days to Adjust = [Recommended Promise] - [Median Promised]

Decisions and Limitations

90% service level was used as a stated reference point rather than a finding.

States below 300 delivered orders were excluded from the analysis (850 orders, 0.9%).

Matrix cells below 100 orders were suppressed.

Review data shows the cost of being late but not the conversion cost of a longer quoted window.

3.1% of customers placed more than one order. The dataset does not support a retention analysis, so that area was dropped.

Thank you!